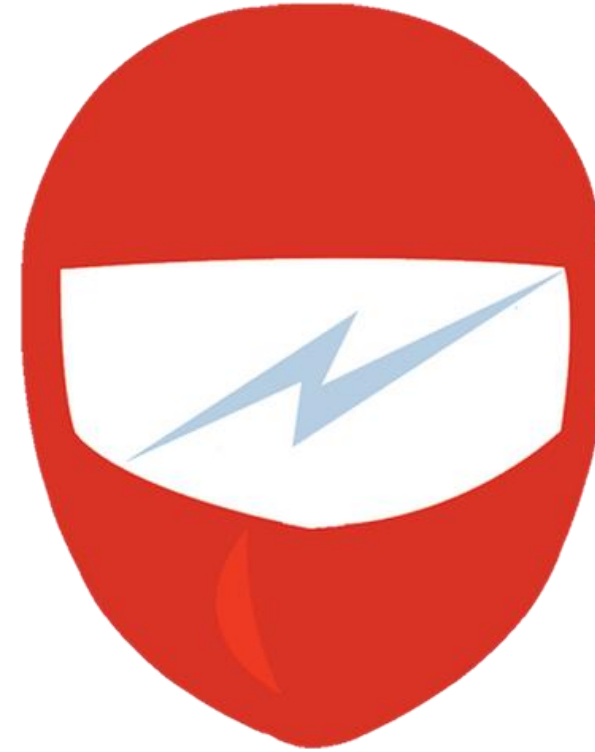




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Who am I ?

Belal Elmikkawy

MTE Level 400

EV subteam Director

Control subteam Director



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RULES

- No side talks
- Phones are in silent mode
- Questions after each slide
- Take permission before leaving
- There will be a break during the session if you want to eat or drink
- Make sure to take notes about what is discussed



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Autonomous System



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Session 7

“Control Theory”



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Agenda

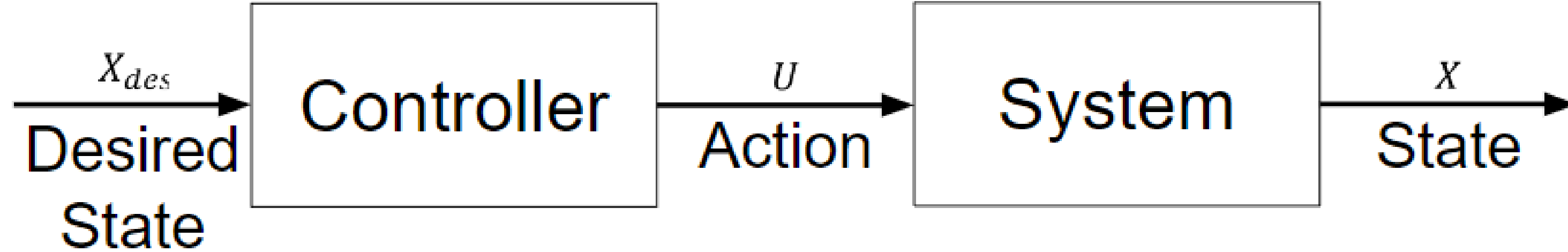
- **Control Theory**
- **PID Controller**
- **PID Issues**
- **PID Modelling**



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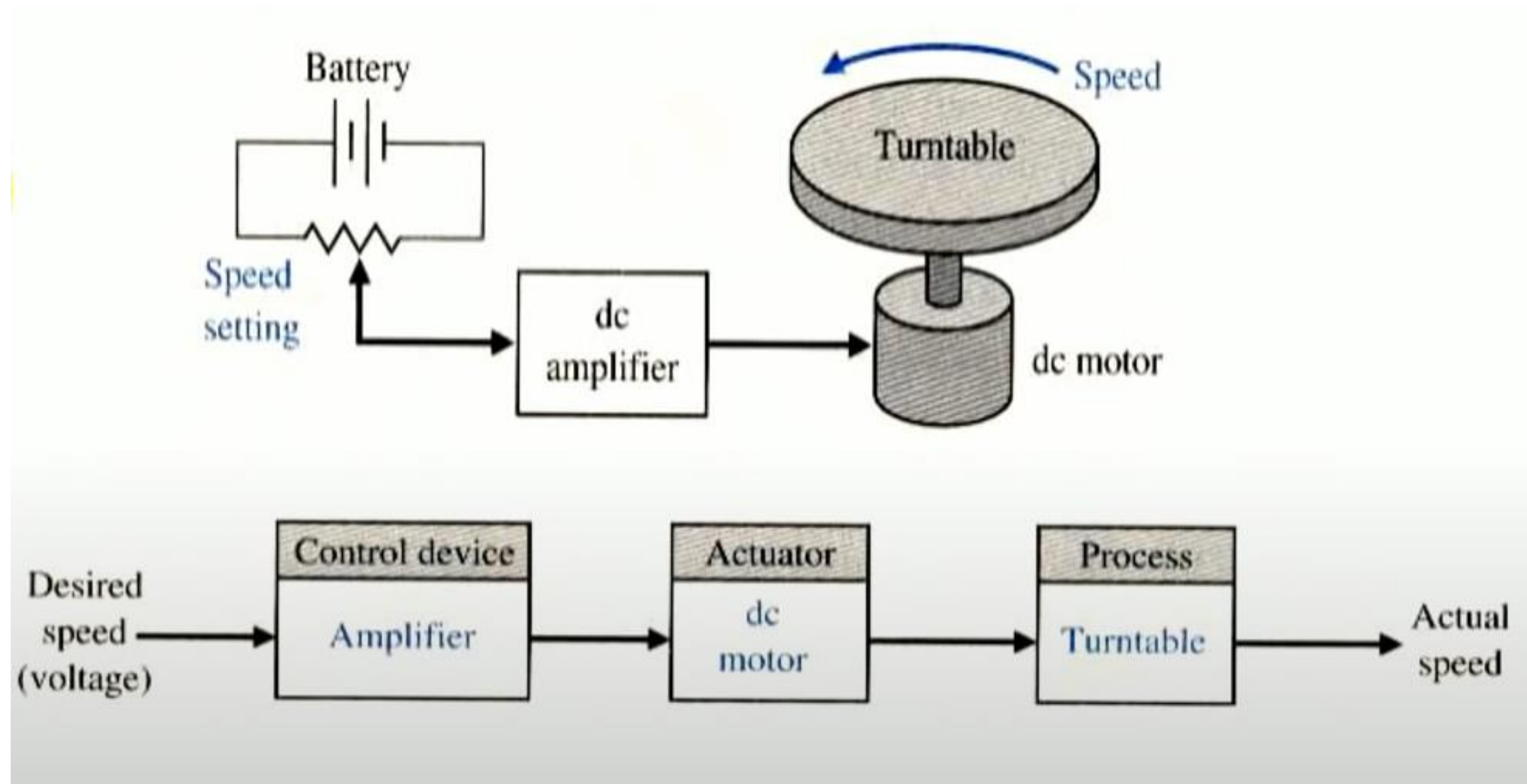
Control Systems

- Open Loop System



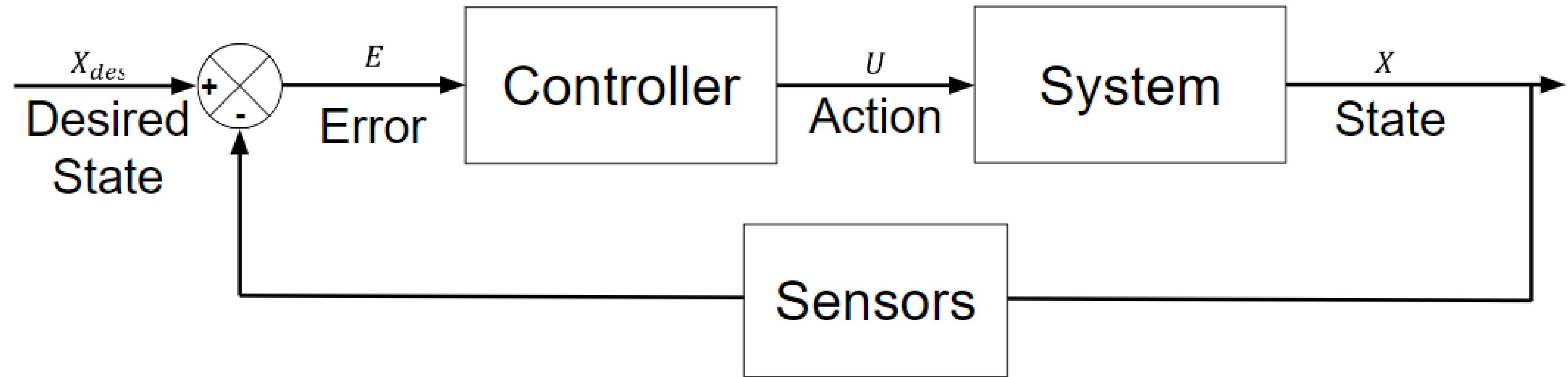
Control Systems

- Open Loop System Example



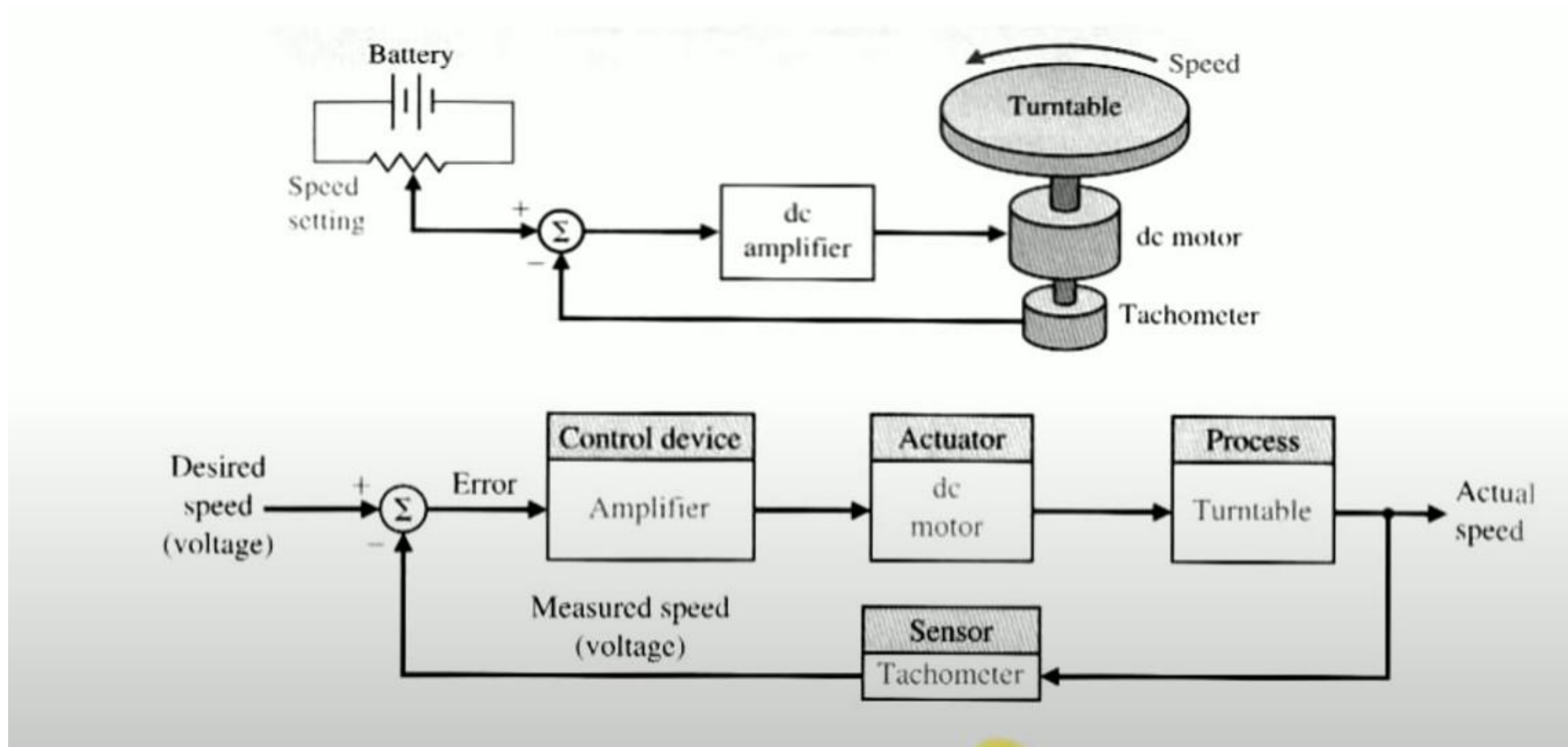
Control Systems

- Closed Loop System

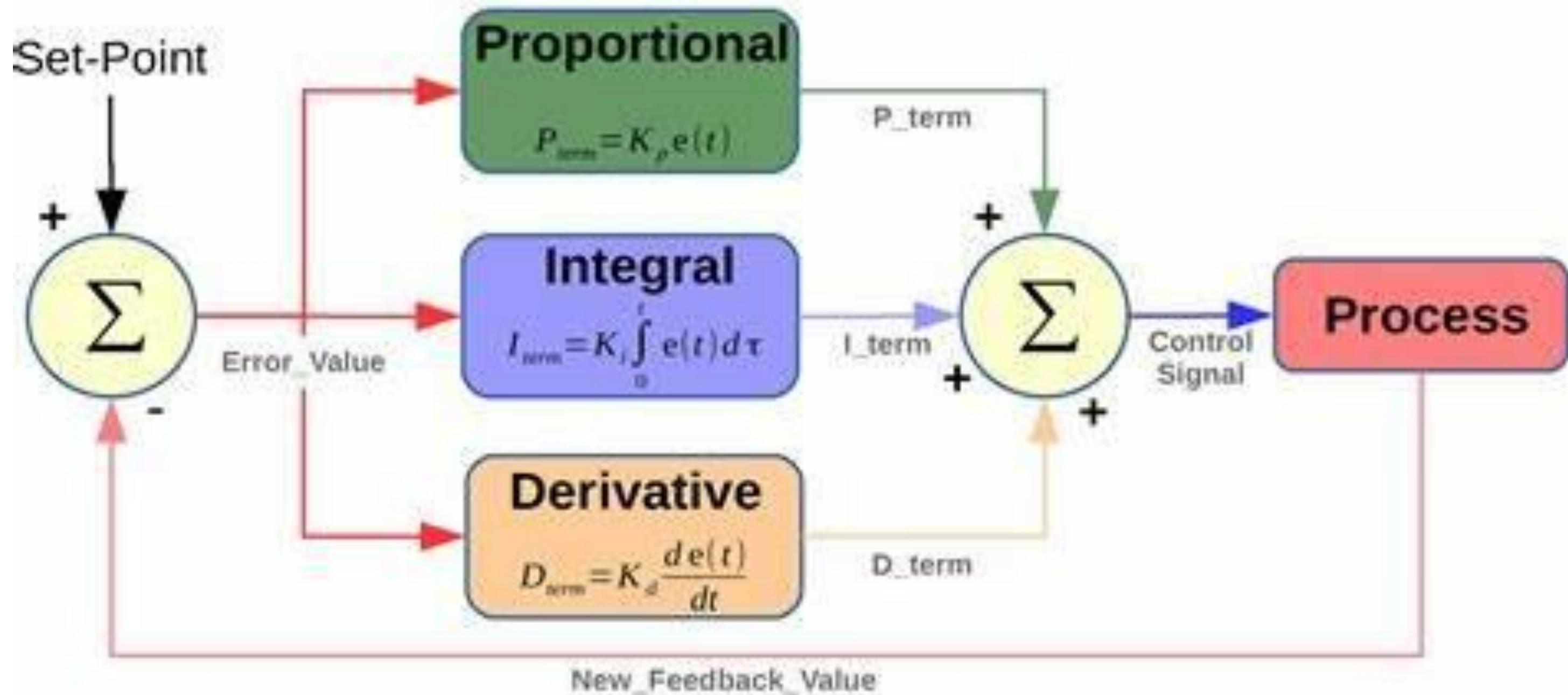


Control Systems

- Closed Loop System Example



PID Controller



PID Controller Parameters

Parameter	Rise Time	Overshoot	Settle Time	S.S.E	Stability
K_p ↑	↓	↑	Small Change	↓	↓
K_i ↑	↓	↑	↑	Eliminate	↓
K_d ↑	Small Change	↓	↓	No Effect	↓



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PID Issues

- **Noise (low pass filter)**
- **Wind-up (clamping-back calculation-observer approach)**



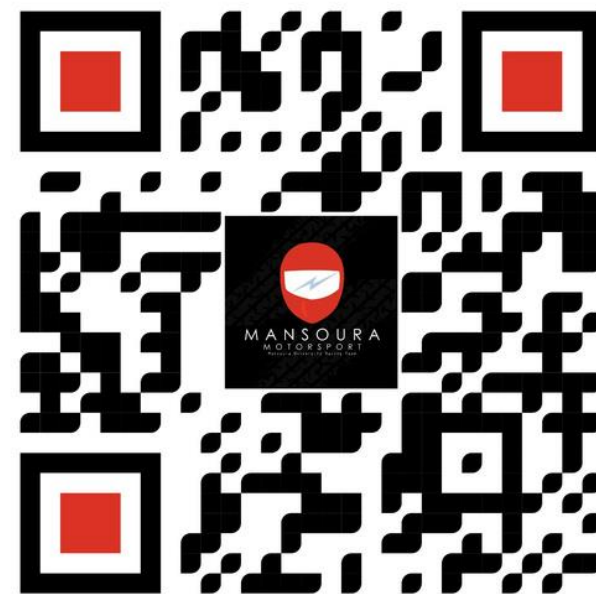
Useful links

- [PID](#)
- [BLDC Matlab Model](#)
- [Control System](#)



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Thank You



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